

BioVyzr

Vyzr Technologies, a Toronto-based company, have created a very unique face-shield called the BioVyzr. The BioVyzr takes protection to the next level, coming equipped with capabilities to protect the wearer's face from not just pathogens, but from water and other hazards.

One look at the BioVyzr and you know what they mean when they say the face shield features a "space-age aesthetic". It almost looks like a space suit. According to Vyzr Tech,



this is a Powered Air Purifying Respirator (PAPR) device. This is something that's typically used only in industrial and healthcare settings and normally pretty pricey at \$1800 a piece. However, the BioVyzr is going for as low as \$172 a piece. For comparison, N95 masks cost about \$40.

The shield is made out of Neoprene and vinyl, and uses anti-fogging material. It has an N95 filter, and a fan that lasts up to 12 hours, purifying air. The fans not only draw air in through the filters, but also send air out of the shield. The seal is pretty airtight, is functional in just about any weather, doubling as a raincoat of sorts. It's also pretty light for its size, at roughly 1.5Kgs, with the battery pack. Also, you can be sure that you aren't touching your face while wearing this face shield. No worries for fit as well, as the BioVyzr comes with adjustable straps.

Theia

Guide dogs are a well-known aid for the blind or visually impaired. However, there are situations where you simply can't have a guide dog. It could be your place of residence, your society or being allergic to dogs etc. To that end, a student, Anthony Camu from Loughborough University, has designed a handheld robotic guide dog, called Theia, to help those who for whatever reason don't have access to guide dogs.

The device basically channels the functions of a guide dog, when held in your hand. They're able to help visually impaired go their way. Inside the Theia you've got a gyroscope, the same kind that's found inside control systems of spacecrafts. It processes real-time online data such as weather and traffic to help users more thoroughly, in both indoor and outdoor environments.



Theia is voice activated, and can provide quick and safe routes to destinations. It then simulates the pulling of a dog and is capable of guiding people through elevators, stairs, and even shops. Additionally, there's also the cost, overall, being much cheaper to maintain than an actual guide dog.

Foresight Haptic Vest

Students from Harvard have developed a wearable device to help those who are visually impaired navigate around more easily. While the purpose of this device and the virtual guide dog above are very similar, the execution is fairly different.

The students started with the launch of their startup, Foresight, and then designed a wearable that looks very similar to a bulletproof vest of sorts. Using computer vision tech and soft robotics, they've designed the wearable so it can connect to the user's smartphone camera. The smartphone is placed on the wearable, right under the user's neck. The camera

basically tells the user where objects are, and this information is relayed to the vest, which then inflates the vest to provide haptic feedback.

Rather than vibration, which is fairly common in devices today, this device uses inflation, which is comparatively more comfortable according to Foresight. The vest inflates in accordance to the distance between the user and an object. It detects objects using a computer vision algorithm.

